Laboratory Requisition — Trebuchet Investigation

Unit: Forces & Motion (Yr 7–8) **Requested by:** _____ **Date needed:** _____

Lesson: Culminating investigation **Number of groups:** _____ **Period / room:** _____

What the students are doing

Students launch a beanbag from a desktop trebuchet with **five different counterweights**, three launches each (**15 launches**, allow 50–60 minutes). They measure how far each launch goes (R) with a tape, and a **PASCO wireless sensor strapped to the trebuchet's arm** records how fast the arm swings. Students work from the *Trebuchet Investigation* booklet.

Equipment required

☐	Item	Per group	Total needed	Notes
☐	Desktop trebuchet + counterweight hanger	1	_____	Pin and sling left exactly as set
☐	Slotted masses	1 set	_____	Five counterweights (see note below)
☐	Beanbag projectile, about 25 g	1 + 1 spare	_____	**Identical** for all groups
☐	PASCO 3-Axis Acceleration sensor (PS-3223)	1	_____	With strap; coin cell checked; ID labelled
☐	Tablet or laptop with SPARKvue	1	_____	Paired with that group's sensor
☐	Measuring tape (8 m+)	1	_____	For the range R
☐	Metre ruler	1	_____	Arm lengths, counterweight drop
☐	Masking tape	1 roll	_____	0 m line + exclusion zone
☐	Counters (or coins)	2	_____	To mark the landing point
☐	Electronic balance	shared	_____	To weigh the projectile
☐	Safety glasses	4	_____	One per student — **mandatory**

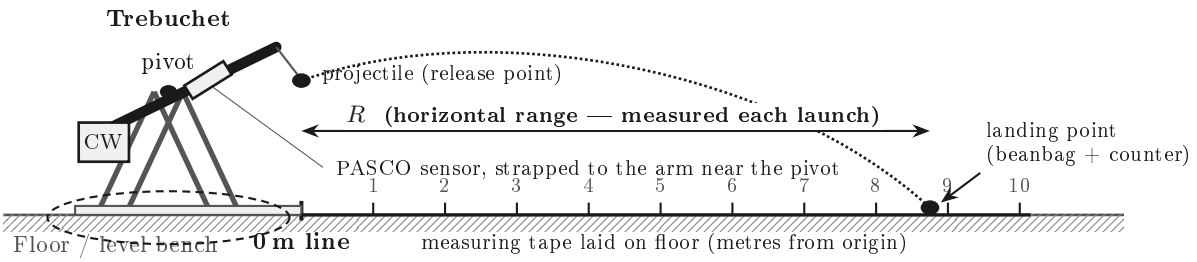
Setup before lesson

1. Clear an 8 m floor strip per station, all launching one way.
2. Tape a **"0 m" line** along the front edge of each base; lay the measuring tape from it down the range.
3. Mark a **1 m exclusion zone** around each trebuchet.
4. Sort the slotted masses into five counterweights per station.
5. Check sensor coin cells; pair each with its group's device.

Safety

- **Risk level: Medium.** Teacher supervises all launches.
- Safety glasses worn by all students during launching.
- Only one person operates each trebuchet; others stand **behind** it.
- Nobody in front of or beside the launch path.
- Beanbags only — no ball bearings or golf balls.
- Projectiles retrieved only on teacher's "all clear".
- Masses changed only with the arm hanging at rest.

Setup diagram (plan for one station)



Key: R = distance from the "0 m" line (front of the base) to where the projectile first lands, measured each launch. **Counterweights:** the lightest that throws reliably to the heaviest the frame takes safely, five evenly spaced.

Notes / queries for teacher: _____ **Prepared by (tech):** _____